

# Ames House Price Prediction

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## 1. Introduction

### Project Objective

The objective is to predict residential house prices using regression models on the Ames Housing dataset. The workflow covers EDA, preprocessing, feature engineering, model training, tuning, evaluation, and interpretation.

### Dataset

The Kaggle `train.csv` dataset contains 1,460 records. `SalePrice` is the target variable. An 80/20 train-test split with `random_state=42` was used.

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## 2. Exploratory Data Analysis

Q1. Why is a skewed target a problem for linear regression, and how does the log transform help?

`SalePrice` was positively skewed (skewness **1.883**, kurtosis **6.536**). A highly skewed target can make errors less balanced and can reduce the effectiveness of linear modelling. After applying `log1p(SalePrice)`, skewness reduced to **0.121**, giving a more symmetric target for modelling.

Q2. Which features look strongest, and which look useless or redundant?

The strongest numerical relationships with `SalePrice` were `OverallQual` (0.791), `GrLivArea` (0.709), `GarageCars` (0.640), `GarageArea` (0.623), `TotalBsmtSF` (0.614), and `1stFlrSF` (0.606). Some features were highly correlated with each other, such as `GarageCars/GarageArea` (0.882), `YearBuilt/GarageYrBlt` (0.826), `GrLivArea/TotRmsAbvGrd` (0.825), and `TotalBsmtSF/1stFlrSF` (0.820), so redundancy was considered during modelling.

### Missing values and outliers

Missing values representing feature absence were treated separately from genuine missing information. Four observations had `GrLivArea > 4000`; they were retained because the values were not automatically invalid.

The dataset contained no duplicate rows and no `YearRemodAdd < YearBuilt` cases.

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## 3. Preprocessing

Q1. Why must imputers and scalers be fitted after the train/test split?

They must be fitted using training data only. Otherwise, information from the test set can enter the preprocessing step and cause data leakage.

Q2. What goes wrong if every NA is blindly filled with the median or mode?

Some NA values in Ames mean that a feature is absent, such as no garage, basement, pool, or fireplace. Replacing these with a median or mode would incorrectly suggest that the feature exists and would lose useful information.

Q3. Why is one-hot encoding a poor fit for ordinal quality ratings?

Ordinal quality categories have a natural order such as Ex > Gd > TA > Fa > Po. One-hot encoding removes this ordering. Therefore, quality variables were mapped to ordered integer values.

Preprocessing applied

- Absence-related columns were filled with "None".
- Genuine gaps such as LotFrontage, MasVnrArea, and Electrical were imputed using training-set statistics.
- Nominal variables were one-hot encoded with unseen-category handling.
- Skewed numerical features were transformed using log1p.
- SalePrice was modelled as log1p(SalePrice) and predictions were converted back using expm1.
- MSSubClass, MoSold, and YrSold were treated as categorical/time-like variables.

4. Feature Engineering

Features created

| Feature               | Purpose                                                  |
|-----------------------|----------------------------------------------------------|
| TotalSF               | Combines basement and above-ground floor area            |
| TotalBath             | Represents total bathroom availability                   |
| HouseAge              | Captures property age at sale                            |
| YearsSinceRemodel     | Captures time since remodelling                          |
| TotalPorchSF          | Combines porch and deck areas                            |
| HasPool               | Indicates presence of a pool                             |
| HasGarage             | Indicates presence of a garage                           |
| HasBasement           | Indicates presence of a basement                         |
| HasFireplace          | Indicates presence of a fireplace                        |
| IsRemodeled           | Indicates whether the property was remodelled            |
| OverallQual_GrLivArea | Captures the interaction between quality and living area |

OverallQual\_GrLivArea had a correlation of approximately 0.825 with SalePrice, compared with 0.786 for OverallQual, showing that the interaction captured additional information. TotalSF had a correlation of approximately 0.323, whileGrLivArea had approximately 0.691; therefore, engineered features were not automatically assumed to replace the raw variables.

Q1. Which engineered feature helped most, and why?

The **OverallQual\_GrLivArea** interaction showed the strongest direct relationship among the engineered features recorded in the analysis. It combines two important house characteristics: quality and living area.

Q2. Why can TotalSF help linear models more than tree models?

A linear model benefits from a single feature that directly represents total usable house area. Tree models can already split across several raw area features, so the additional handcrafted combination may provide less extra information.

Feature selection

Mutual information was used as a feature-selection analysis. The highest scores included **QualxArea**, **OverallQual\_GrLivArea**, **TotalSF**, **OverallQual**, and **GrLivArea**.

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## 5. Modeling

All models used the same train/test split and the log-transformed target.

Models evaluated: - Linear Regression - Ridge Regression - Lasso Regression - SVR (RBF) - KNN Regression - Decision Tree Regression - Random Forest Regression - XGBoost Regression

Five-fold cross-validation was used for hyperparameter tuning. Random Forest and XGBoost tuning produced the following results:

Model Best 5-Fold CV RMSE

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Random Forest 0.143189 XGBoost 0.128097

Q1. Why do SVR and KNN fail without feature scaling, while trees do not care?

SVR and KNN depend on distances or feature magnitudes, so variables with larger numerical ranges can dominate the model. Tree-based models make split decisions based on thresholds, so scaling does not affect their split ordering.

Q2. What does the training-to-cross-validation gap tell you?

A large gap indicates that a model performs much better on training data than on unseen validation data, which is evidence of overfitting. A small gap generally indicates better generalization, although both training and validation errors must be considered.

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## 6. Evaluation and Comparison

Model comparison

Model Comparison

| Model            | RMSE Log | RMSE USD  | MAE USD   | R <sup>2</sup> |
|------------------|----------|-----------|-----------|----------------|
| Lasso Regression | 0.124382 | 25,721.42 | 15,127.89 | 0.917095       |

| Model             | RMSE Log | RMSE USD  | MAE USD   | R <sup>2</sup> |
|-------------------|----------|-----------|-----------|----------------|
| Linear Regression | 0.124559 | 23,532.54 | 15,182.97 | 0.927802       |
| Ridge Regression  | 0.129875 | 26,371.04 | 15,773.23 | 0.909611       |
| XGBoost           | 0.133438 | 26,203.78 | 16,304.35 | 0.904583       |
| Random Forest     | 0.150580 | 30,127.16 | 18,123.19 | 0.878493       |
| Decision Tree     | 0.209495 | 37,619.74 | 25,693.71 | 0.764815       |
| SVR (RBF)         | 0.242815 | 53,452.43 | 31,107.02 | 0.684053       |
| KNN Regression    | 0.290186 | 59,842.50 | 36,990.33 | 0.548753       |

Q1. Which model wins, and is the margin over the runner-up larger than the CV standard deviation?

Using the primary metric, log-scale RMSE, Lasso had the lowest recorded value (**0.124382**), followed closely by Linear Regression (**0.124559**). The margin is approximately **0.000177**. The CV standard deviation for these two models was not recorded in the current results table, so the margin cannot be compared reliably with their CV standard deviation.

Q2. Where do residuals fail and why?

The residual analysis was performed using the best and worst models. Large errors are expected where house characteristics are unusual or where the available features do not fully capture the factors affecting sale price. Expensive or unusual properties can therefore produce larger residuals.

Extra experiment

The requested raw-versus-log/outlier comparison was not completed in the current workflow.

## 7. Interpretation and Justification

Q1. Do Random Forest, XGBoost, and Linear Regression feature rankings agree?

The rankings are not exactly the same, but some important features are common. Random Forest gave the highest importance to **OverallQual\_GrLivArea**, while XGBoost gave high importance to **OverallQual** and **OverallQual\_GrLivArea**. Linear Regression had **GrLivArea** among its important features.

So, the models show some agreement about house quality and living area, but their exact feature rankings are different.

Q2. What do the 10 worst predictions have in common?

The largest prediction error was for a house with an actual price of \$755,000, where the model predicted about \$203,644. Other large errors were also seen for high-priced houses.

This shows that the Lasso model had more difficulty predicting some expensive houses accurately.

### Q3. Bias-variance profile of the models

*Linear Regression:*\* It has a simple structure, so it usually has lower variance but may have higher bias for nonlinear relationships. In this experiment, it performed well with an RMSE Log of 0.124559.

**SVR:** SVR can capture nonlinear patterns but is sensitive to scaling and parameter choices. Its RMSE Log of 0.242815 shows weaker performance in this experiment.

**KNN:** KNN depends strongly on nearby training points and can be sensitive to feature scaling. It had the highest RMSE Log of 0.290186, showing weaker generalization.

**Decision Tree:** A single tree can learn complex patterns but can also overfit, resulting in high variance. Its RMSE Log was 0.209495.

**Random Forest:** Random Forest uses many trees and averages their predictions, which helps reduce variance compared with a single tree. It achieved an RMSE Log of 0.150580.

**XGBoost:** XGBoost builds trees sequentially and can learn complex patterns, but it can also overfit if not tuned properly. Its RMSE Log was 0.133438, showing strong performance in this experiment.

### Q4. Which model would you deploy and why?

For deployment, I would consider both model performance and practical use. Lasso Regression gave the lowest RMSE Log of 0.124382, while Linear Regression gave the lowest RMSE in USD and the highest  $R^2$ .

Since the primary evaluation metric is RMSE on log-transformed SalePrice, Lasso Regression would be my main candidate. However, Linear Regression is also a strong option because it has good accuracy and is simple to interpret and maintain.

### Q5. Three things to try next

1. Try wider hyperparameter tuning for Random Forest and XGBoost to improve their performance.
2. Test additional feature engineering and remove features that add little useful information.
3. Try ensemble or stacking methods to combine the strengths of different models.

### Q6. Dataset limitation

The dataset represents houses from Ames, Iowa, during a specific time period. Therefore, the model may not generalize well to other cities, regions, or current housing markets.

The dataset also has a limited number of observations, so more diverse and recent housing data would help improve the model's generalization.

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## 8. Conclusion

The project demonstrated an end-to-end regression workflow for Ames house-price prediction. Log transformation, missing-value handling, categorical encoding, feature engineering, and model comparison were used to build and evaluate several regression approaches.

The recorded results showed very close performance between Lasso and Linear Regression on the primary log-scale metric, while Linear Regression produced the lowest RMSE in USD and highest  $R^2$ . The results also

showed that model selection should consider the evaluation metric, interpretability, and practical deployment requirements rather than relying on a single score.

## 9. Reproducibility

- Train/test split: 80/20
- Random seed: 42
- Target: `log1p(SalePrice)`
- Prediction conversion: `expm1`
- Python environment versions: listed in `requirements.txt`